SHUBHAM PATHNEJA

Ex-Robotics Intern @IIT MANDI | Team Lead @SAARTHI AI | ROS | Computer Vision | Robotics | Kinematics | Control Systems

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LinkedIn | Github | Portfolio

Tools - VS Code | Gazebo | MuJoCo | Jetson Edge Devices | Rviz | Ubuntu |

Programming Tools - Python | C++

Technologies - ROS/ROS2 | Computer Vision | Control Systems | VLA | Path Planning | Kinematics | TensorRT | CUDA | Pytorch | Simulation |

Soft Skills : Team Work | Leadership | Time Management | Communication | Problem Solving | Project Planning |

EXPERIENCE & PROJECTS

Robotics Intern, IIT MANDI (CAIR) (*Certificate*) *May 2026 – July 2026*

- **Description:** Developed and implemented end-to-end autonomous navigation and perception algorithm on the Clearpath Husky A200 mobile robot at IIT Mandi's Centre of AI and Robotics. Integrated GPS tracking using a host website and dense path following logic system., using Autoware's models and NVIDIA's TensorRT and CUDA on Jetson AGX Orin.
- Developed a obstacle avoidance fused with GPS path tracking algorithm for the Last Mile Delivery Problem.

Unexplored Area Navigation ***(Ongoing)*** *July 2026 - Present*

- **Description:** Architected an autonomous exploration and SLAM pipeline for the exploring of the unknown workspace for the Trossen Robotics LoCoBot WX250 6DOF under the guidance of Dr. Raja Rout in Robotics Lab, Thapar University.
- Integrated RGB-D camera spatial perception and 2D LiDAR data for real-time obstacle avoidance, dynamic path planning, and active mapping in unknown environments.
- **Tech stack and Frameworks:** ROS2 Galactic, SLAM, Rviz, Nav2, Control Systems

Marketplace Navigation Robot Simulator (*Github*) *March 2026 - June 2026*

- **Description:** Designed and simulated a differential-drive autonomous mobile robot (AMR) in MuJoCo for the indoor supermarket navigation and product location assistance for large supermarkets. The system utilizes 2D LiDAR-based SLAM for environment mapping and Nav2 for global/local path planning.
- **Tech stack and Frameworks:** ROS2 Humble, SLAM, Rviz, Nav2, MuJoCo

SAARTHI (Smart AI assistant for Task Handling and Information) ***(Ongoing)*** *June 2025 - Present*

- **Description:** Engineering an intelligent AI task-scheduling assistant that processes user tasks via NLP, applies a custom weighted scoring model using Reinforcement Learning to capture the user's behavior, and assigns tasks to optimal time slots according to user flexibility.
- **My Contribution:** Lead the team for the product and implemented the human behavior tracking reinforcement learning algorithm along with the voice to command generation functionality.
- **Tech stack and Frameworks:** Python, Decision Trees, NLP, Flutter, FastAPI, CP-SAT, Reinforcement Learning, PostgreSQL

SafeSight(*Link*) *January 2026 – April 2026*

- **Description:** Built a real-time behavioral anomaly detection system on existing CCTV infrastructure using Faster-RCNN and ResNet50, achieving 80.71 % detection accuracy across 3 object classes on a live deployment.
- **My Contribution:** Lead the team and planned the pipeline along with implementing the ResNet model for fast inference merging it with the backend and the website.
- **Tech stack and Frameworks:** Computer Vision, Flask app, Faster-RCNN with ResNet50, Reactjs

EDUCATION

Bachelors of Engineering in Robotics and Artificial Intelligence *(2023 - 2027)*
Thapar Institute of Engineering and Technology, Patiala, Punjab

- **CGPA -** 8.12 / 10.00
- **Societies** - Lead as the Head of Designing in Fine Arts and Photography Society, TIET
  Served as a core member of Thapar Venture Club, TIET